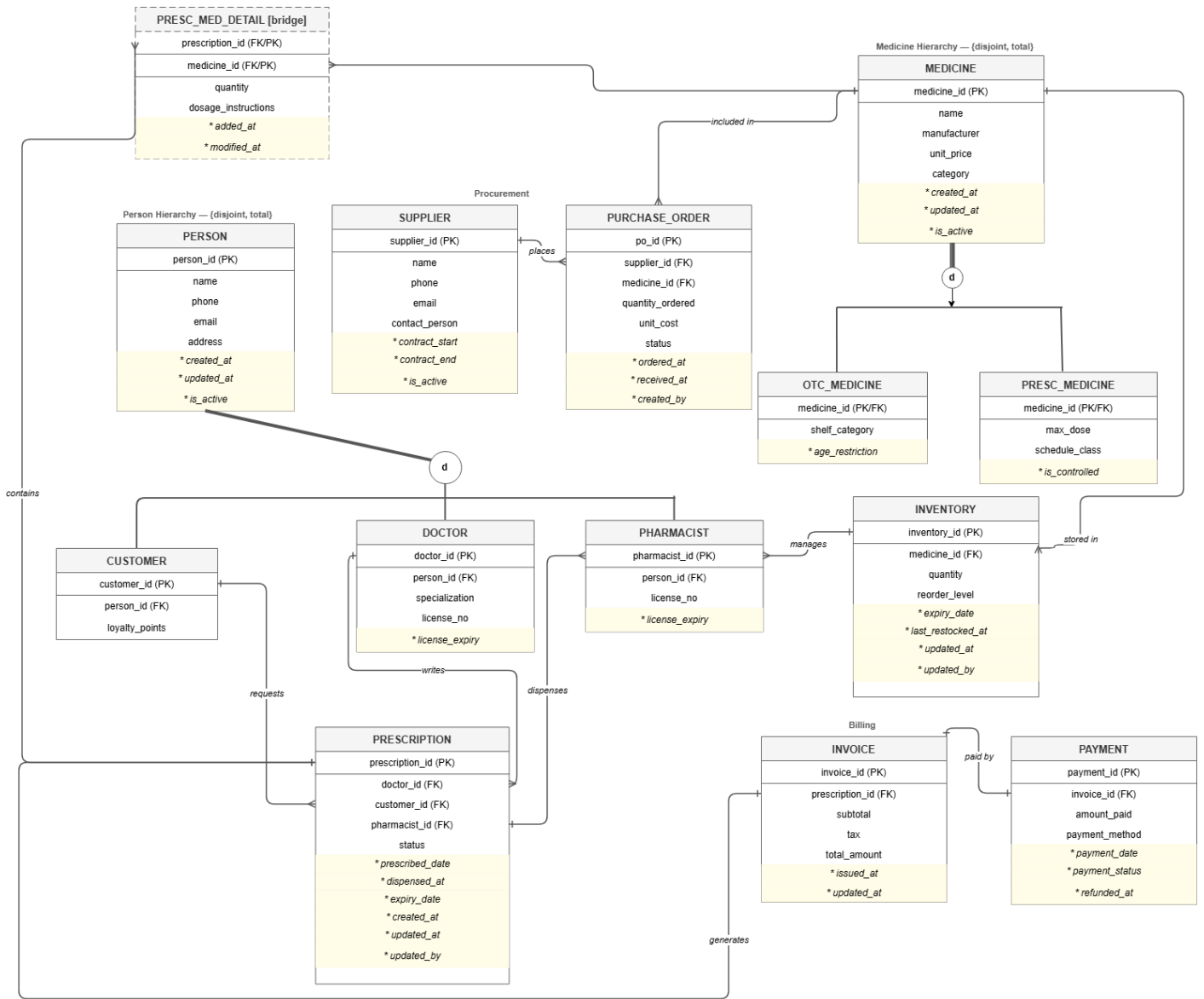


EER DIAGRAM:



YELLOW+*: REQUIRED.

PERSON:

```
-- PERSON (Super)
CREATE TABLE PERSON (
  person_id SERIAL PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  phone VARCHAR(20) NOT NULL,
  email VARCHAR(100) NULL,
  address VARCHAR(200) NULL,
  created_at TIMESTAMP DEFAULT NOW(),
  updated_at TIMESTAMP NULL,
  is_active BOOLEAN DEFAULT TRUE
);
-- DOCTOR(sub)
CREATE TABLE DOCTOR (
  doctor_id SERIAL PRIMARY KEY,
  person_id INT NOT NULL REFERENCES PERSON(person_id),
  specialization VARCHAR(100) NOT NULL,
  license_no VARCHAR(50) NOT NULL,
  license_expiry DATE NOT NULL
);
-- PHARMACIST(sub)
CREATE TABLE PHARMACIST (
  pharmacist_id SERIAL PRIMARY KEY,
  person_id INT NOT NULL REFERENCES PERSON(person_id),
  license_no VARCHAR(50) NOT NULL,
  license_expiry DATE NOT NULL
);
-- CUSTOMER(sub)
CREATE TABLE CUSTOMER (
  customer_id SERIAL PRIMARY KEY,
  person_id INT NOT NULL REFERENCES PERSON(person_id),
  loyalty_points INT DEFAULT 0
);
```

MEDICINE:

```
--MEDICINE
CREATE TABLE MEDICINE (
  medicine_id SERIAL PRIMARY KEY,
  name VARCHAR(150) NOT NULL,
  manufacturer VARCHAR(150) NOT NULL,
  unit_price NUMERIC(10,2) NOT NULL,
  category VARCHAR(100) NOT NULL,
  created_at TIMESTAMP DEFAULT NOW(),
  updated_at TIMESTAMP NULL,
  is_active BOOLEAN DEFAULT TRUE
);
--OTC_MEDICINE
CREATE TABLE OTC_MEDICINE (
  medicine_id INT PRIMARY KEY REFERENCES MEDICINE(medicine_id),
  shelf_category VARCHAR(100) NOT NULL,
  age_restriction VARCHAR(50) NULL
);
--PRESC_MEDICINE
CREATE TABLE PRESC_MEDICINE (
  medicine_id INT PRIMARY KEY REFERENCES MEDICINE(medicine_id),
  max_dose VARCHAR(50) NOT NULL,
  schedule_class VARCHAR(50) NOT NULL,
  is_controlled BOOLEAN DEFAULT FALSE
);
```

PERCRIPTION:

```
-- PRESCRIPTION
CREATE TABLE PRESCRIPTION (
  prescription_id SERIAL PRIMARY KEY,
  doctor_id INT NOT NULL REFERENCES DOCTOR(doctor_id),
  customer_id INT NOT NULL REFERENCES CUSTOMER(customer_id),
  pharmacist_id INT NULL REFERENCES PHARMACIST(pharmacist_id),
  status VARCHAR(30) DEFAULT 'Pending',
  prescribed_date DATE NOT NULL,
  expiry_date DATE NOT NULL,
  dispensed_at TIMESTAMP NULL,
  created_at TIMESTAMP DEFAULT NOW(),
  updated_at TIMESTAMP NULL,
  updated_by VARCHAR(100) NULL
);

-- PRESC_MED_DETAIL
CREATE TABLE PRESC_MED_DETAIL (
  prescription_id INT NOT NULL REFERENCES PRESCRIPTION(prescription_id),
  medicine_id INT NOT NULL REFERENCES MEDICINE(medicine_id),
  quantity INT NOT NULL,
  dosage_instructions VARCHAR(300) NOT NULL,
  added_at TIMESTAMP DEFAULT NOW(),
  modified_at TIMESTAMP NULL,
  PRIMARY KEY (prescription_id, medicine_id)
);
```

Supplier and Purchase order

```
-- SUPPLIER
CREATE TABLE SUPPLIER (
  supplier_id SERIAL PRIMARY KEY,
  name VARCHAR(150) NOT NULL,
  phone VARCHAR(20) NOT NULL,
  email VARCHAR(100) NULL,
  contact_person VARCHAR(100) NOT NULL,
  contract_start DATE NOT NULL,
  contract_end DATE NULL,
  is_active BOOLEAN DEFAULT TRUE
);

--PURCHASE ORDER
CREATE TABLE PURCHASE_ORDER (
  po_id SERIAL PRIMARY KEY,
  supplier_id INT NOT NULL REFERENCES SUPPLIER(supplier_id),
  medicine_id INT NOT NULL REFERENCES MEDICINE(medicine_id),
  quantity_ordered INT NOT NULL,
  unit_cost NUMERIC(10,2) NOT NULL,
  status VARCHAR(30) DEFAULT 'Ordered',
  ordered_at TIMESTAMP DEFAULT NOW(),
  received_at TIMESTAMP NULL,
  created_by VARCHAR(100) NOT NULL
);
```

INVENTORY:

```
--| INVENTORY
CREATE TABLE INVENTORY (
    inventory_id SERIAL PRIMARY KEY,
    medicine_id INT NOT NULL REFERENCES MEDICINE(medicine_id),
    quantity INT NOT NULL DEFAULT 0,
    monitor_level INT NOT NULL DEFAULT 10,
    expiry_date DATE NULL,
    last_restocked TIMESTAMP NULL,
    updated_at TIMESTAMP DEFAULT NOW(),
    updated_by VARCHAR(100) NOT NULL
);

-- INVOICE
CREATE TABLE INVOICE (
    invoice_id SERIAL PRIMARY KEY,
    prescription_id INT NOT NULL REFERENCES PRESCRIPTION(prescription_id),
    subtotal NUMERIC(10,2) NOT NULL,
    tax NUMERIC(10,2) DEFAULT 0,
    total_amount NUMERIC(10,2) NOT NULL,
    issued_at TIMESTAMP DEFAULT NOW(),
    updated_at TIMESTAMP NULL
);

-- PAYMENT
CREATE TABLE PAYMENT (
    payment_id SERIAL PRIMARY KEY,
    invoice_id INT NOT NULL REFERENCES INVOICE(invoice_id),
    amount_paid NUMERIC(10,2) NOT NULL,
    payment_method VARCHAR(50) NOT NULL,
    payment_date TIMESTAMP DEFAULT NOW(),
    payment_status VARCHAR(30) DEFAULT 'Completed',
    refunded_at TIMESTAMP NULL
);
```

Entries:

```
-- PERSON
INSERT INTO PERSON (name, phone, email, address, is_active) VALUES ('Dr. Salman Raza', '0321-1001001', 'salman@bui.edu.pk', 'F-8/1, Islamabad', TRUE);

-- DOCTOR
INSERT INTO DOCTOR (person_id, specialization, license_no, license_expiry) VALUES (1, 'General Physician', 'PMC-001122', '2026-12-31');

-- PHARMACIST
INSERT INTO PHARMACIST (person_id, license_no, license_expiry) VALUES (4, 'PPC-441144', '2026-03-31');

-- CUSTOMER
INSERT INTO CUSTOMER (person_id, loyalty_points) VALUES (6, 150);

-- MEDICINE
INSERT INTO MEDICINE (name, manufacturer, unit_price, category, is_active) VALUES ('Panadol 500mg', 'GSK Pakistan', 15.00, 'Analgesic', TRUE);

-- OTC_MEDICINE
INSERT INTO OTC_MEDICINE (medicine_id, shelf_category, age_restriction) VALUES (1, 'Pain Relief', NULL);

-- PRESC_MEDICINE
INSERT INTO PRESC_MEDICINE (medicine_id, max_dose, schedule_class, is_controlled) VALUES (2, '3 times/day', 'Schedule-H', FALSE);

-- SUPPLIER
INSERT INTO SUPPLIER (name, phone, email, contact_person, contract_start, contract_end, is_active) VALUES ('MedCo Distributors', '051-1112223', 'medco@dist.pk', 'Arif Butt', '2024-01-01', '2025-12-31', TRUE);

-- PURCHASE_ORDER
INSERT INTO PURCHASE_ORDER (supplier_id, medicine_id, quantity_ordered, unit_cost, status, ordered_at, received_at, created_by) VALUES (1, 1, 500, 12.00, 'Received', '2025-01-10', '2025-01-15', 'Hina Baig');

-- INVENTORY
INSERT INTO INVENTORY (medicine_id, quantity, monitor_level, expiry_date, last_restocked, updated_by) VALUES (1, 480, 50, '2027-06-01', '2025-01-15', 'Hina Baig');

-- PRESCRIPTION
INSERT INTO PRESCRIPTION (doctor_id, customer_id, pharmacist_id, status, prescribed_date, expiry_date, dispensed_at, updated_by) VALUES (1, 1, 1, 'Dispensed', '2025-01-05', '2025-04-05', '2025-01-06 10:30', 'Hina Baig');

-- PRESC_MED_DETAIL
INSERT INTO PRESC_MED_DETAIL (prescription_id, medicine_id, quantity, dosage_instructions) VALUES (1, 2, 1, 'Take 1 tablet 3 times a day after meals for 7 days');

-- INVOICE
INSERT INTO INVOICE (prescription_id, subtotal, tax, total_amount, issued_at) VALUES (1, 225.00, 0, 225.00, '2025-01-06');

-- PAYMENT
INSERT INTO PAYMENT (invoice_id, amount_paid, payment_method, payment_date, payment_status) VALUES (1, 225.00, 'Cash', '2025-01-06', 'Completed');
```